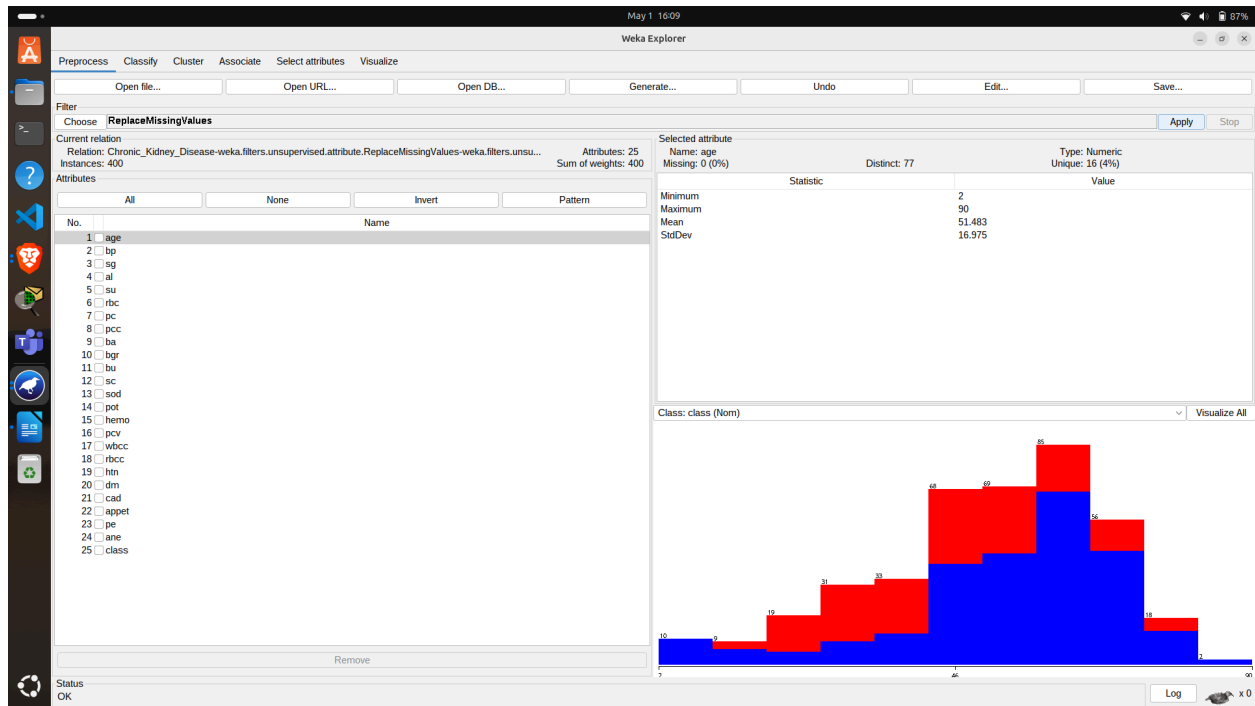
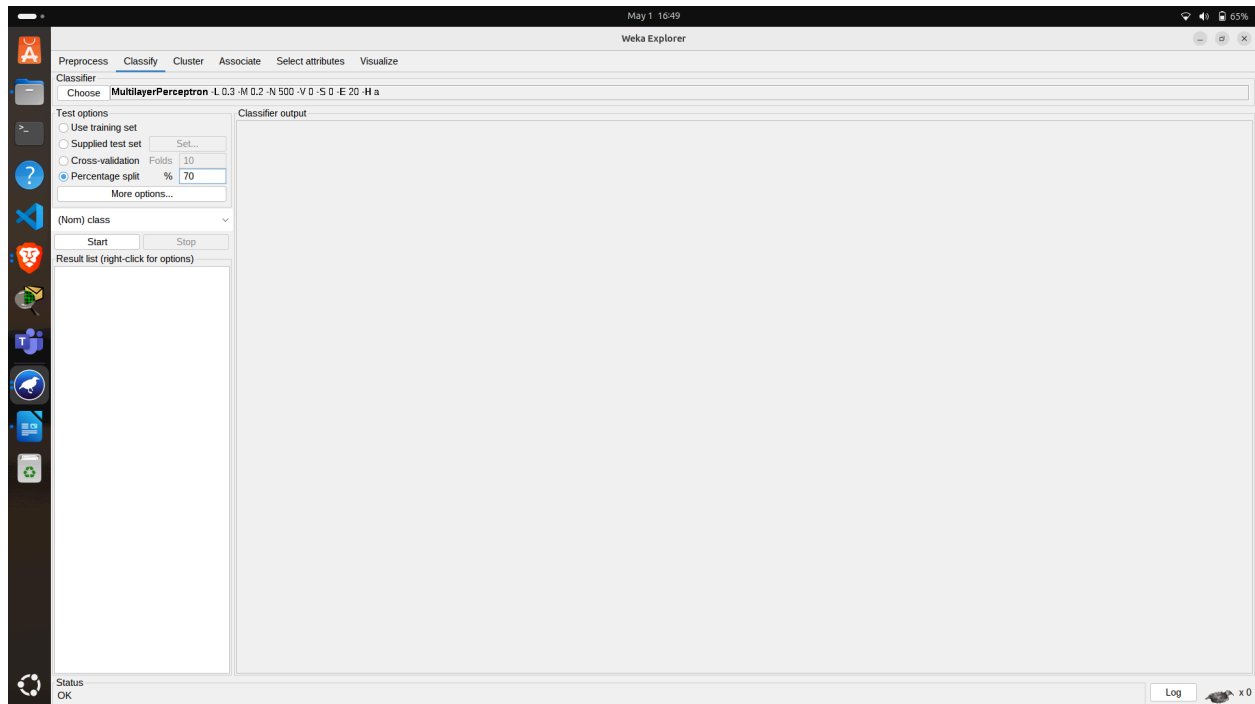


Simon Mwaura -Neural Network – MLP



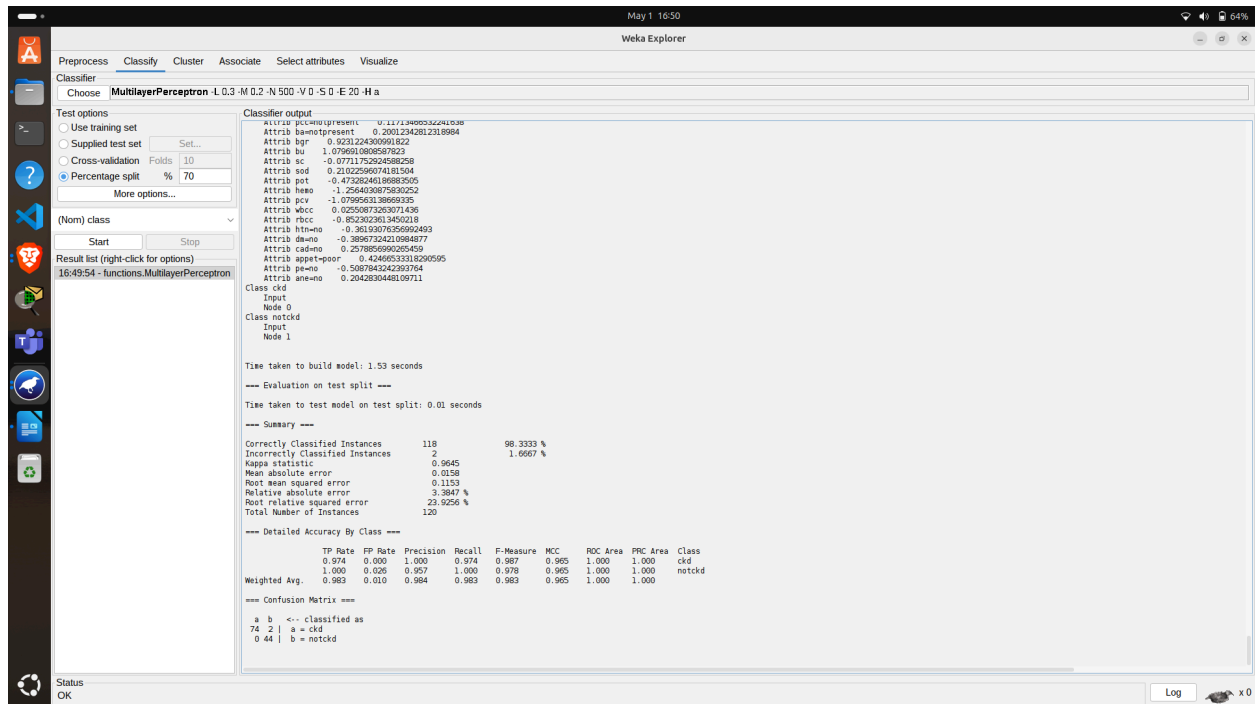
Screenshot 1: Data Preprocessing & EDA

This screenshot demonstrates the initial Exploratory Data Analysis (EDA) and preprocessing phase using the WEKA Explorer. The ReplaceMissingValues filter has been successfully applied to the dataset of 400 instances to handle the missing data, which is a necessary step before training a neural network. The age attribute is currently selected, showing 0% missing values. The histogram provides a visual distribution of patient ages against the target class (red for CKD, blue for not-CKD), revealing that CKD cases in this dataset peak heavily in the 50–70 age range.



Screenshot 2: MLP Algorithm Setup

This screenshot shows the configuration of the Neural Network model in the Classify tab. The MultilayerPerceptron classifier has been selected. Per the assignment instructions, the evaluation method is explicitly set to a **Percentage split of 70%**. This ensures that 70% of the data (280 instances) will be used to train the network, reserving the remaining 30% (120 instances) as unseen data to test its predictive accuracy.



Screenshot 4: Final Evaluation & Confusion Matrix.

Accuracy: The MLP achieved a highly accurate classification rate of **98.33%** (118 out of 120 instances correctly classified).

Confusion Matrix Observations: The confusion matrix at the bottom reveals exactly where the model succeeded and failed. It correctly identified 74 CKD patients and 44 healthy patients. It produced 0 False Positives (no healthy patients were falsely flagged). However, it produced **2 False Negatives** (2 patients with actual CKD were incorrectly classified as healthy).

Conclusion: While 98.33% is an excellent accuracy score, in a medical context, those 2 False Negatives represent a significant clinical risk, as a sick patient would go undiagnosed.